FACTS-MAS: A Multi-Agent Architecture for Adaptive Forecasting from Heterogeneous Evidence

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Abstract

Real-world forecasting often requires integrating heterogeneous evidence whose predictive value varies across horizons, operating regimes, and domains. We introduce FACTS-MAS, a domain-agnostic multi-agent forecasting framework that decomposes evidence into specialized channels for temporal dynamics, exogenous context, seasonality, structural information, and discrete events. Each agent generates an independently testable forecast through a common interface, while leakage-safe historical validation measures its predictive skill relative to a naive reference. These skill estimates are used to weight and combine agent forecasts, so that uninformative evidence is either removed or downweighted rather than forced to participate. A calibration procedure selects domain-specific constant means based on historical validation folds. We evaluate FACTS-MAS on weekly for-sale housing inventory from Zillow and weekly unemployment insurance claims from FRED, for 15 U.S. metropolitan areas at 4-, 8-, and 13-week forecasting horizons. Our method outperforms both a naive last-value forecast method and the TimesFM foundation model. However, a factor-adjusted exponential smoothing model performs as well as the fused forecast. The evidence therefore supports FACTS-MAS as a systematic protocol for testing, calibrating, and recomposing evidence channels, before assembling them to the strongest single model for a specific application domain. Code, data, and results are available at https://github.com/Madumita24/FACTS-MAS.

1 Introduction

Forecasting in operational settings almost always draws on more than one kind of information. A target series may be strongly autoregressive, yet its near-term path can also depend on exogenous conditions, recurring seasonal cycles, slowly varying attributes of the entity being forecast, and discrete events that do not arrive as ordinary numerical features. These information sources differ in representation, timing, and, most importantly, predictive value. A signal that helps at a short horizon or in one regime may add nothing at a longer horizon or in another domain.

A common approach in forecasting is to fold all available inputs into a single statistical/machine-learning (ML) model or a single artificial intelligence (AI) reasoning process. This forecasting approach may be able to capture complex interactions among different information sources, but it binds three tasks into one object: how each source is encoded, how the forecast is produced, and how credit is assigned to each source. When the forecast accuracy has changed, it is difficult to identify whether the change came from temporal dynamics, external context, structure, or events. It also requires dissimilar information sources to be translated into a single input format, even though time-series dynamics, calendar effects, static covariates, and unstructured events are not naturally expressed in the same way.

In this paper, we develop a new approach, Forecasting from Adaptive, Calibrated, Testable Sources – Multi-agent AI System (FACTS-MAS), which is built upon the opposite premise. We define AI agents by

their *functional evidence roles*, not by an application name. Each agent turns the evidence in its channel into a numerical forecast that satisfies a common output contract. The fusion step then asks a focused question: *does this channel improve on a simple reference forecast in leakage-safe historical validation?* We quantify the relative skill of each agent for the targeted forecasting task as a weight; weak or redundant channels are downweighted rather than forced to participate. Domain-specific constants like event-impact scales, event durations, and reversion caps are chosen by the same validation test, and these constants can be set to zero. This changes the interpretation of negative results. If an agent receives little weight, it is not necessarily a failed component; instead, it is evidence that the channel it is managing contains little incremental information for the current domain, or that the present implementation does not extract the useful information. Likewise, a plausible causal event may in fact have little predictive power in one domain, but we can keep it and assign it a confidence score for later domains.

We evaluate FACTS-MAS on the forecast of weekly U.S. housing inventory, a setting in which all five candidate channels can be populated: recent inventory dynamics, national financial conditions, annual listing seasonality, metropolitan structure, and disaster declarations. The study covers 15 metropolitan statistical areas (MSAs) at forecast horizons of 4, 8, and 13 weeks. At each forecast origin, defined as the week in which a prediction is issued, models are trained only on observations dated on or before that week; because Zillow publishes each week's value one to two weeks later, we also report every result with a two-week publication lag. Evaluation uses a six-fold expanding-window backtest, with a 13-week embargo between the end of training and the start of testing, followed by a single pre-registered holdout on data after February 21, 2026 that was not accessed until the configuration was frozen. On Zillow's smoothed inventory series, the frozen configuration reduces MAPE relative to a naive last-value forecast from 4.194% to 2.118% at the 4-week horizon, from 7.225% to 3.622% at 8 weeks, and from 10.814% to 5.731% at 13 weeks in the backtest, and from 3.122% to 1.243%, 5.445% to 1.929%, and 8.438% to 3.045% on the holdout. The reduction is smaller on the unsmoothed series (24% at 4 weeks in the backtest, against 50% on the smoothed series). The fused forecast is significantly more accurate than the TimesFM foundation model [Das et al., 2024] in the backtest. On the holdout it has a lower error than a chain-of-thought language-model forecast built on the NEXUS prompt [Das et al., 2026], but the difference is not statistically significant. It is not more accurate than a factor-adjusted damped exponential smoothing model, with which it is statistically equivalent at 8 weeks. We also transfer the method to weekly unemployment insurance claims to assess the generalizability of the approach.

The contributions of this paper are as follows.

1. We introduce FACTS-MAS, a multi-agent forecasting procedure that assigns heterogeneous evidence to functional channels—temporal, exogenous, seasonal, structural, and event-based—and requires each agent to return a numerical forecast under a common output contract.

2. We develop a leakage-safe fusion rule that measures each agent's improvement over a naive reference, aggregates that skill by the median, and converts the resulting scores into fusion weights. The weights may depend on forecast horizon and operating regime when the validation sample is sufficient to support that conditioning; a regime is used only when it contains at least 13 distinct validation weeks, and otherwise the pooled weights apply.

3. We specify a calibration procedure for domain-specific constants, including event-impact scales, event durations, and reversion caps. Candidate values are selected on historical validation folds, and a constant may be set to zero when validation does not support a nonzero effect.

4. We evaluate the method on weekly for-sale housing inventory for 15 U.S. metropolitan areas at forecast horizons of 4, 8, and 13 weeks, on both Zillow's smoothed and unsmoothed series. The analysis compares accuracy with naive, seasonal-naive, and seasonally adjusted naive references, with exponential-

smoothing and Theta models, with the TimesFM foundation model, and with a chain-of-thought language-model forecast, using panel Diebold–Mariano tests and block-bootstrap confidence intervals. It distinguishes evidence channels that add incremental skill from channels whose quantitative effect is weak or unsupported.

5. We report a pre-registered evaluation: the configuration, the hypotheses, and the statistical tests were fixed and committed before a holdout period of 29 weekly origins was accessed, and every result on that holdout is reported, including those that do not favor the fused forecast.

The remainder of the paper is organized as follows. Section 2 reviews multi-agent forecasting, the evolution of forecasting models, forecast combination, and the NEXUS precursor. Section 3 defines the FACTS-MAS procedure, including the evidence channels, the shared forecast interface, skill-based fusion, calibration, and attribution. Section 4 states the rolling-origin protocol, the information boundary, the accuracy metrics, and the comparison design, then instantiates the method for U.S. housing inventory and reports forecast accuracy together with the channel and fusion diagnostics. Section 5 discusses the housing findings, including the holdout results, and the transfer of the method to weekly unemployment insurance claims. Section 6 concludes.

2 Related Work

FACTS-MAS draws on four lines of research: multi-agent forecasting, the succession of time-series model classes, the combination of heterogeneous evidence, and LLM-assisted interpretation of unstructured information. This section reviews each line and states how FACTS-MAS differs.

2.1 Multi-Agent and LLM-Assisted Forecasting

Multi-agent systems decompose a complex prediction task into specialized components whose outputs are orchestrated or aggregated [Chang et al., 2025, Wang et al., 2024]. Large language models have widened this pattern: agents can interpret text, call tools, plan, and communicate over mixed structured and unstructured inputs [Yao et al., 2023, Jang et al., 2026]. Recent agentic forecasting systems organize those capabilities into workflows—planning, tool use, iterative revision, and ensemble generation—rather than a single-pass map from history to the future [Tao et al., 2026, Pan et al., 2026]. Role-specialized designs give agents distinct duties and pass intermediate results through conversation or a shared workflow [Park et al., 2023, Wu et al., 2024, Hong et al., 2024].

However, specialization alone does not establish that every agent adds predictive information. Two agents may encode the same signal, a plausible feature family may have no out-of-sample benefit, and usefulness may depend on the forecast horizon or the operating regime. When agents only converse, their individual numerical contributions cannot be isolated. FACTS-MAS differs by requiring each agent to issue its own numerical forecast and by measuring that forecast's skill relative to a simple reference before fusion assigns it any weight. The skill measurement may depend on the forecast horizon and the operating regime when the validation sample is sufficient.

2.2 From Statistical Models to Foundation Models

Time-series forecasting has moved through several model classes that remain useful as channel implementations or as baselines. Exponential smoothing, including the Holt–Winters trend form, forecasts from a local level and trend [Holt, 2004, Winters, 1960, Hyndman et al., 2002, Pan, 2011]. Additive statistical models of the same family are widely used in practice [Taylor and Letham, 2018]. Large competitions then compared

these statistical methods with later machine-learning entries on common collections of series [Makridakis and Hibon, 2000, Makridakis et al., 2020]. The M4 winner combined exponential smoothing with a recurrent network [Smyl, 2020]. Global neural forecasters fit one model to many related series, including autoregressive recurrent networks [Salinas et al., 2020], residual basis expansions [Oreshkin et al., 2020], and attention models that also accept covariates [Lim et al., 2021]. Lim and Zohren [2021] survey this deep-learning line.

Transformer forecasters adapted attention to long horizons [Zhou et al., 2021, Wu et al., 2021, Nie et al., 2023], and Wen et al. [2023] survey that design family. Subsequent comparisons showed that simple linear models can match or beat several of those architectures [Zeng et al., 2023]. Pretrained foundation models now produce zero-shot forecasts across domains, including TimesFM [Das et al., 2024], Chronos [Ansari et al., 2024], and Moirai [Woo et al., 2024]. Those zero-shot forecasts can still carry variance and systematic bias. Bootstrap aggregation, regression stacking, and residual correction of a foundation-model forecast have reduced both on an operational electricity-load series [Modi and Pan, 2025]. A parallel line asks a language model to forecast, either by reprogramming a frozen model [Jin et al., 2024] or by encoding the series as text [Gruver et al., 2023]. FACTS-MAS does not replace this sequence of forecasters. A channel may use a statistical smoother, a regression, or a pretrained model. The housing study uses Holt's linear-trend exponential smoothing on the temporal channel at the 4-week horizon and a factor-adjusted damped-trend model [Gardner and McKenzie, 1985] at 8 and 13 weeks, and it uses TimesFM 2.5, a later checkpoint of the TimesFM line, as a numerical baseline.

2.3 Heterogeneous Evidence and Forecast Combination

Operational forecasting problems commonly pool evidence with heterogeneous statistical structure: endogenous dynamics, exogenous covariates, calendar effects, slowly varying entity attributes, and discrete events. Recurrent events can be forecast through the cumulative intensity of the event process, including with boosted trees that use both static and dynamic features [Liu and Pan, 2021]. A joint statistical or machine-learning model maps this collection to a single forecast, so that input encoding, prediction, and source-level attribution are determined by one fitted mapping. Forecast combination retains a separate forecast for each source and aggregates those forecasts. Equal-weight aggregation reduces dependence on any one specification [Clemen, 1989]. Performance-based weights assign greater influence to component forecasts with lower historical error [Bates and Granger, 1969]. Regression on the component forecasts is another classical route to those weights [Granger and Ramanathan, 1984, Stock and Watson, 2004]. Static weights assume that relative accuracy is sufficiently stable to be pooled across time [Newbold and Granger, 1974, Wang et al., 2023]. Timmermann [2006] reviews when such combination improves on the best individual forecast. That question is central here: a combination of channels is only useful if it improves on the strongest single model that could have been selected on the same validation data. A related prediction task is degradation under a time-varying environment, where an ensemble is trained to follow that change [Wang et al., 2020].

FACTS-MAS uses rolling weights based on each agent's skill relative to a simple reference forecast. The skill score, its median aggregation, and the softmax conversion are defined in Section 3. Whether those weights should also depend on forecast horizon or operating regime is treated there as a validation choice, because finer conditioning is estimated from fewer observations.

2.4 Calibration and NEXUS

Plausible numerical assumptions—event-severity maps, effect durations, lags, and thresholds—can degrade forecast accuracy even when the underlying causal mechanism is plausible. FACTS-MAS selects such

constants on historical validation folds and allows a constant to be set to zero when a nonzero effect is unsupported. The calibration rule is given in Section 3.

The immediate precursor of our MAS approach is NEXUS [Das et al., 2026], which forecasts from mixed structured and unstructured evidence by passing that evidence through multi-agent language-model reasoning and then issuing the forecast. Efficiency extensions of that design, such as early-exit gating and batched reasoning, retain language-model reasoning on the main forecasting path; they are not part of FACTS-MAS and are not evaluated here. FACTS-MAS assigns the same evidence to specialized channels and obtains an independent numerical forecast from each channel. The fusion weights are estimated from the channels' historical skill relative to a simple reference on a validation window that ends at the training cutoff, and they are re-estimated whenever the training window is extended. Those weights may also depend on the forecast horizon and the operating regime when the validation sample is sufficient.

The housing experiments in Section 4.2 compare FACTS-MAS with a chain-of-thought language-model forecast that uses the NEXUS prompt, with a time-series foundation model, and with strong statistical references. Those comparisons test the forecasting consequence of this decomposition.

3 FACTS-MAS: Domain-Agnostic Forecasting Architecture

FACTS-MAS separates a reusable forecasting procedure from any one domain's data and models. The appropriate raw variables, predictive models, event types, and operating regimes may depend on the domain, but the forecasting procedure remains the same. Heterogeneous evidence is digested and fed into forecasting channels; each channel produces an independently evaluable forecast, and these forecasts are combined adaptively.

Let y_t denote the target variable observed up to forecast origin t, and let H denote the requested forecast horizon. Let $\mathcal{I}_t$ denote the information available at that origin. For a forecasting domain $\mathcal{D}$, this information is streamed into K functional evidence channels,

$$\mathcal{E}^{(\mathcal{D})} = \left\{ E_1^{(\mathcal{D})}, E_2^{(\mathcal{D})}, \ldots, E_K^{(\mathcal{D})} \right\}. \tag{1}$$

$E_k^{(\mathcal{D})}$ denotes the channel, and each channel is assigned one agent. The assigned agent reads that channel and maps the available information to a horizon forecast,

$$f_k : \mathcal{I}_t \to \hat{\mathbf{y}}_k, \qquad k = 1, \ldots, K, \tag{2}$$

where $\hat{\mathbf{y}}_k$ is the forecast over the requested horizon. In a named domain $\mathcal{D}$, the agent is A_k,

$$A_k : \mathcal{I}_t^{(\mathcal{D})} \to \hat{\mathbf{y}}_{k,t}^{(H)}, \tag{3}$$

where $\hat{\mathbf{y}}_{k,t}^{(H)} = [\hat{y}_{k,t+1}, \ldots, \hat{y}_{k,t+H}]$ is the H-step forecast produced independently by agent k.

These maps determine the forecast. At each forecast origin, domain evidence is assigned to the K channels, and each agent returns an independent forecast. A validation step scores each agent's historical forecasts against a reference, the resulting skill scores set the fusion weights, and adaptive fusion combines the agent forecasts into $\hat{\mathbf{y}}_{\mathrm{FACTS}}$. Calibration is a separate, offline step that selects channel constants on historical validation folds before the test period begins.

Figure 1 shows this procedure at one forecast origin. The following subsections describe the components of this process and identify which elements remain invariant across domains and which need to be reconfigured for specific forecasting tasks.

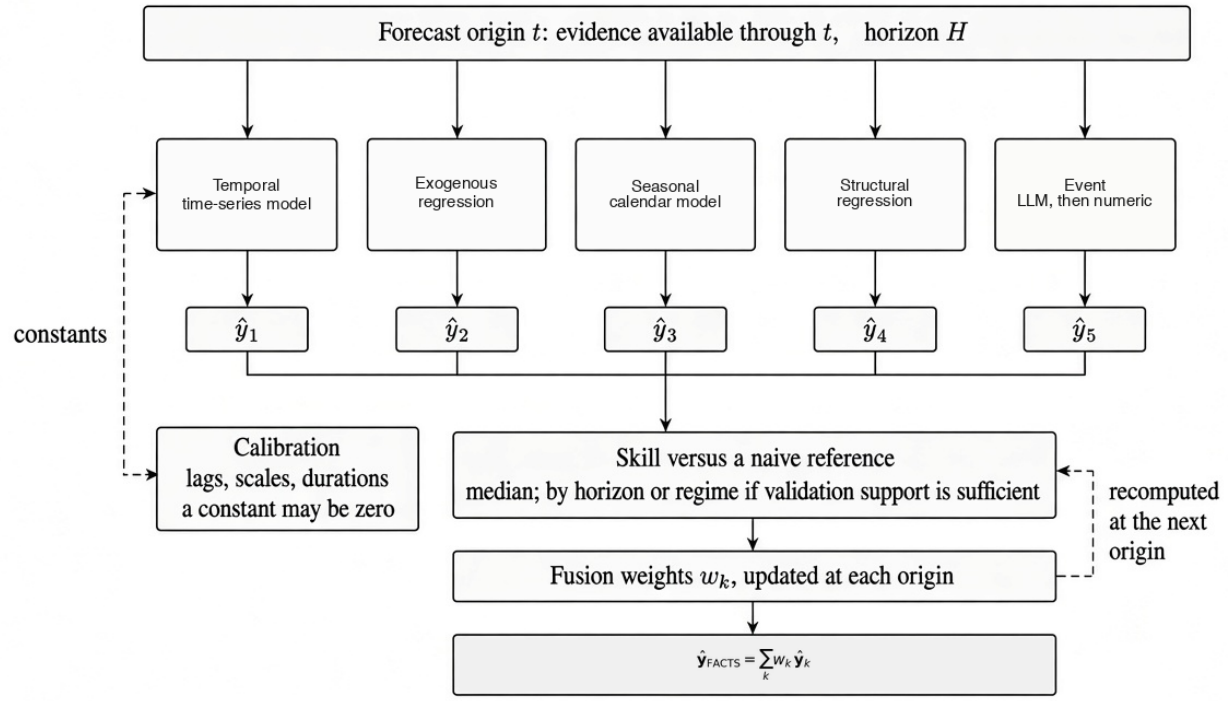


Figure 1: FACTS-MAS at one forecast origin. Each evidence channel returns an independent horizon forecast. Skill relative to a naive reference, measured on a validation window that ends at the training cutoff, sets the fusion weights. Calibration acts offline on channel constants and may set a constant to zero.

3.1 Functional Evidence-Channel Decomposition

FACTS-MAS organizes forecasting evidence according to its functional role rather than according to a particular dataset or application. The current architecture defines five candidate evidence channels:

1. **Temporal channel.** It models the endogenous dynamics of the target using its historical trajectory. This channel captures stability, trend, autocorrelation, and other temporal structure that can be estimated without external information.

2. **Exogenous channel.** It represents external numerical variables that may influence future changes in the target. The identity and transmission lag of these variables are domain dependent.

3. **Seasonal channel.** It represents recurring temporal structure associated with calendar, periodic, or cyclic behavior.

4. **Intrinsic or structural channel.** It represents relatively persistent characteristics of the entity being forecast. These features may distinguish entities even when they share similar recent temporal trajectories.

5. **Event channel.** It represents discrete or unstructured information whose forecasting relevance may require semantic interpretation before it can be converted into a structured signal.

These channels are *functional abstractions*, not fixed feature sets. For example, the exogenous channel may contain monetary-policy indicators in a housing application, weather and demand drivers in an energy

application, or labor-market indicators in an employment application. Similarly, the meaning of a relevant event depends on the target domain.

This distinction is central to domain transfer. FACTS-MAS does not assume $E_k^{(\mathcal{D}_1)} = E_k^{(\mathcal{D}_2)}$. Instead, it assumes that both domains can expose evidence serving comparable forecasting functions, $E_k^{(\mathcal{D}_1)} \sim E_k^{(\mathcal{D}_2)}$ with respect to functional role, while allowing their data sources, features, transformations, and agent implementations to differ.

3.2 Shared Agent Interface

To keep evidence-channel implementation separate from downstream fusion, every forecasting agent must satisfy a common output contract. For a given forecast request, an agent returns

$$O_k = (a_k, d, t, H, \hat{\mathbf{y}}_k, \mathbf{c}_k)\,, \tag{4}$$

where a_k identifies the agent, d identifies the entity or forecasting unit, t is the forecast origin, H is the requested horizon, $\hat{\mathbf{y}}_k$ contains exactly H forecast values, and $\mathbf{c}_k$ optionally contains horizon-specific confidence values.

This output contract enforces three important invariants. First, all agents must produce forecasts for the same entity, origin, and requested horizon before fusion. Second, the output length must exactly match the requested horizon. Third, target forecasts must satisfy domain-valid output constraints; for the housing implementation, for example, inventory forecasts are constrained to be non-negative.

The shared contract allows an agent implementation to be replaceable without modifying the fusion layer. A statistical model, machine-learning model, foundation time-series model, or LLM-assisted semantic embedding model can therefore occupy an evidence channel as long as its output satisfies the same forecasting interface.

3.3 Leakage-Safe Agent Evaluation

The quality of a forecasting agent is estimated using its historical forecasts generated under the same information constraints that would exist during deployment. For each forecast origin t, an agent may access only evidence available on or before that origin after accounting for publication and availability delays.

For an agent k and reference model b, define the absolute percentage errors at validation point i as

$$e_{k,i} = \frac{|y_i - \hat{y}_{k,i}|}{|y_i|}, \qquad e_{b,i} = \frac{|y_i - \hat{y}_{b,i}|}{|y_i|}. \tag{5}$$

FACTS-MAS measures incremental forecasting skill relative to the reference, $s_{k,i} = 1 - e_{k,i}/e_{b,i}$. Thus,

$$s_{k,i} \begin{cases} > 0, & \text{agent improves on the reference,} \\ \approx 0, & \text{agent approximately matches the reference,} \\ < 0, & \text{agent underperforms the reference.} \end{cases} \tag{6}$$

Skill is scored against the persistence baseline, so an agent that only reproduces that baseline receives a skill near zero.

Directly averaging the pointwise skill ratios can be unstable when the reference error is extremely small at individual validation points. FACTS-MAS therefore aggregates pointwise skill using the median,

$$\tilde{s}_k = \mathrm{median}_{i \in \mathcal{V}}\,(s_{k,i})\,, \tag{7}$$

where $\mathcal{V}$ denotes the leakage-safe validation window. The median reduces the influence of isolated points at which an unusually small reference error would otherwise produce an extreme ratio.

3.4 Skill-Based Adaptive Fusion

The validated skill scores are converted into non-negative normalized weights using a temperature-controlled softmax,

$$w_k = \frac{\exp(\tilde{s}_k/\tau)}{\sum_{j=1}^{K} \exp(\tilde{s}_j/\tau)}, \tag{8}$$

where $\tau > 0$ controls how strongly differences in historical skill affect the final allocation. In the housing configuration $\tau = 0.05$. This value was set by hand from the magnitude of validation skill scores, not selected by a search; a validation-selected temperature was tested and not adopted (Section 5.1.5).

The fused forecast is then

$$\hat{\mathbf{y}}^{(H)}_{\mathrm{FACTS},t} = \sum_{k=1}^{K} w^{(H)}_{k,t} \hat{\mathbf{y}}^{(H)}_{k,t}. \tag{9}$$

Softmax weighting is useful in this setting because it preserves the complete ranking of agents, including agents with negative relative skill. No hard threshold is required to assign identical minimum weights to all underperforming channels. An agent that is slightly worse than the reference can therefore remain distinguishable from an agent that is substantially worse.

Weights are re-estimated whenever the training window is extended rather than fixed once for the entire evaluation period. In the backtest, each fold estimates its weights from a 104-week validation window ending at the training cutoff and holds them fixed over that fold's test window. Consequently, $w^{(H)}_{k,t_1} \neq w^{(H)}_{k,t_2}$ is permitted across folds, and within a fold when t_1 and t_2 fall in different regimes.

3.5 Adaptive Context and Regime Conditioning

The relative value of an evidence channel may depend on both forecast horizon and operating context. FACTS-MAS therefore supports multiple levels of conditioning.

Let r_t denote a domain-specific regime observed at forecast origin t. The regime variable itself is not part of the invariant framework. It is a domain-specific configuration satisfying two requirements: it must represent a meaningful state for the target process, and its value at forecast origin t must be computable without future information. The same historical skill scores can be grouped in four ways. Pooled weighting, w_k, uses the full validation history and assigns each agent one weight at every horizon and in every regime. Horizon-conditioned weighting, $w^{(H)}_k$, recomputes skill at each forecast horizon. Regime-conditioned weighting, $w^{(r)}_k$, recomputes skill within each regime. Horizon–regime-conditioned weighting, $w^{(H,r)}_k$, recomputes skill inside each combination of horizon and regime. FACTS-MAS treats the choice among these groupings as a validation problem: for each eligible configuration, competing fusion strategies are evaluated on historical validation data, and the strategy with lower validation error is selected before entering the test window.

This creates an important distinction between *adaptivity* and *granularity*. A more granular rule can represent more context-specific behavior but is estimated from fewer observations. Rare regimes can therefore produce insufficient evidence for reliable weight selection. FACTS-MAS allows the level of adaptation to be constrained by historical support rather than assuming that finer conditioning always improves generalization. In the housing configuration, a regime-specific weight vector is used only when the regime contains at least 13 distinct validation weeks; otherwise the forecast falls back to the pooled weights. An earlier version of the implementation had no such rule and assigned equal weights to every channel when a regime had no validation data, which affected 780 test cells and inflated the apparent benefit of regime conditioning.

3.6 Calibration of Domain-Specific Parameters

Some evidence channels require domain-specific constants or mappings. Examples include event-effect duration, numerical impact scales, classification thresholds, lag lengths, or transformation parameters. Assigning such values from intuition alone can introduce an unmeasured source of forecasting error.

FACTS-MAS therefore separates parameter calibration from held-out evaluation. For a domain-specific parameter θ and candidate set $\Theta = \{\theta_1, \ldots, \theta_m\}$, each candidate is evaluated on historical validation data for validation fold f,

$$L_f(\theta) = \mathrm{MAPE}\left(\mathbf{y}_f, \hat{\mathbf{y}}_f(\theta)\right). \tag{10}$$

A candidate is not promoted solely because it achieves the lowest pooled error. The framework additionally checks whether the direction of improvement is sufficiently consistent across folds. This discourages parameter changes whose apparent benefit is driven by a small subset of historical periods. Most importantly, calibration is permitted to return a null quantitative effect. If $\theta^* = 0$ provides the best and most consistent validation performance, the framework interprets this as evidence that the corresponding magnitude adjustment is unsupported. Other outputs from the same channel, such as classification, confidence, or attribution information, can remain available even when its direct numerical adjustment is removed.

In the housing study, each constant was selected once using the validation windows of all six folds, with a candidate promoted only if it improved validation loss in at least five of the six folds. Because later folds' validation windows postdate earlier folds' test windows, this selection is not fully nested in time. The only adopted calibration result, a zero event-impact scale, changes test accuracy by at most 0.002 percentage points.

3.7 Factor Attribution

Because every agent produces an independent forecast and every fusion weight is derived from historical predictive skill, FACTS-MAS exposes a direct forecast-level attribution mechanism.

For forecast origin t, the contribution of agent k to the forecasted change from the last observed value is

$$C_{k,t}^{(H)} = w_{k,t}^{(H)} \left(\hat{\mathbf{y}}_{k,t}^{(H)} - y_t \mathbf{1}\right), \tag{11}$$

so that $\sum_k C_{k,t}^{(H)} = \hat{\mathbf{y}}_{\mathrm{FACTS},t}^{(H)} - y_t \mathbf{1}$. A level-based share $w_{k,t}^{(H)} \hat{\mathbf{y}}_{k,t}^{(H)}$ would credit an agent that merely repeats the last observed value with a share of the forecast level even though it contributes nothing to the forecasted change; the change-based form assigns such an agent a contribution of zero. The set $\left\{w_{1,t}^{(H)}, \ldots, w_{K,t}^{(H)}\right\}$ provides an interpretable summary of which evidence channels were historically most useful under the validation conditions governing that forecast.

These weights should not be interpreted as causal effect estimates. They are empirical measures of relative predictive contribution under the selected forecasting and validation procedure. The distinction is important: an agent can receive low weight because its evidence is redundant with another channel, because its implementation fails to extract the available signal, or because the evidence itself has little predictive value.

3.8 Domain Adaptation

The domain-agnostic claim of FACTS-MAS concerns the forecasting procedure, not the invariance of every fitted component. Moving from source domain $\mathcal{D}_1$ to target domain $\mathcal{D}_2$ therefore involves a controlled reconfiguration. Table 1 separates the framework into invariant and domain-specific components.

Under this formulation, adapting FACTS-MAS to a new domain consists of four steps:

Table 1: Invariant and domain-specific components of FACTS-MAS.

Invariant framework components	**Domain-specific components**
Evidence-channel decomposition	Raw data sources and feature definitions
Shared agent output contract	Model selected within each evidence channel
Forecast-origin information boundary	Relevant exogenous variables and lags
Leakage-safe validation	Seasonal period or temporal structure
Skill-relative agent evaluation	Structural/entity-level variables
Adaptive fusion mechanism	Event ontology and event sources
Calibration protocol	Calibrated parameters
Factor-attribution procedure	Operating-regime definition

1. **Map available evidence to functional channels.** Identify which historical, exogenous, seasonal, structural, and event information is meaningful for the new forecasting target.

2. **Instantiate the channel models.** Select or fit an appropriate forecasting implementation for each available evidence channel while preserving the shared output contract.

3. **Calibrate domain-specific assumptions.** Estimate lags, scales, thresholds, or other configurable quantities using historical validation data without accessing the held-out evaluation period.

4. **Re-estimate attribution.** Compute agent skill and adaptive fusion weights from the target domain's own historical performance rather than transferring weights learned in another domain.

Consequently, FACTS-MAS does not require $w_k^{(\mathcal{D}_1)} = w_k^{(\mathcal{D}_2)}$. Indeed, different weight distributions are expected if the importance of evidence channels differs between domains. Cross-domain transfer is successful when the common decomposition, validation, calibration, and fusion procedure can be applied without redesigning the overall architecture, while the system learns a domain-appropriate configuration from the available evidence.

To summarize, at each forecast origin, available information is streamed into the functional evidence channels, and each assigned agent returns an independent horizon forecast. A validation step scores these forecasts against a reference, a calibration step sets the domain-specific constants, and the resulting skill scores are converted into fusion weights. Adaptive fusion then combines the channel forecasts into the FACTS-MAS forecast and records each channel's estimated contribution.

4 Case Studies

Two cases—weekly U.S. housing inventory and weekly unemployment insurance claims—are analyzed by FACTS-MAS, and a third, energy demand, is specified as a planned configuration only. They are used to answer the following questions: First, does the fused forecast improve on simple and competitive references? Second, which evidence channels add incremental skill, and does that skill change with horizon or regime? Third, can the same decomposition and fusion procedure be re-instantiated when the target and the raw evidence change? The housing study answers the first two questions, and unemployment claims are used for the third; the energy configuration is reported as a channel map without results. All experiments follow one rolling-origin protocol. Data and channel models may change by domain; the information boundary, agent-level scoring, weight estimation, and held-out testing rules do not.

4.1 Data Preparation and Agent Setups

Observations are ordered as $\mathcal{T} = \{t_1, \ldots, t_N\}$. Each fold f splits this history into training, validation, an optional embargo, and a test region, $\mathcal{T} = \mathcal{T}^{(f)}_{\text{train}} \cup \mathcal{T}^{(f)}_{\text{val}} \cup \mathcal{T}^{(f)}_{\text{embargo}} \cup \mathcal{T}^{(f)}_{\text{test}}$. The split is a rolling-origin evaluation: training expands, forecasts are issued at successive origins, and accuracy is judged on held-out periods rather than on a single origin [Tashman, 2000]. Randomly mixing times into training and test folds can leak future observations in an autoregressive series [Bergmeir et al., 2018], so model fitting, fusion weights, and the choice of weighting rule use only pre-test information. Two exceptions in the development of the housing configuration are disclosed and corrected. The seasonal smoothing window was originally chosen on the test window of the last fold, including targets in January and February 2026; it was re-selected on validation data alone, nested by fold, and the reported results use the re-selected window. The calibrated constants were selected across all folds' validation windows (Section 3.6). Forecasts are issued at every eligible origin in the test window, and every method is scored on that full set. An input x may enter a forecast at origin t only when its availability date satisfies $t_{\text{available}}(x) \le t$, after any publication lag. The embargo period takes at least the longest horizon, $H_{\max} = \max\{H_1, \ldots, H_q\}$; for weekly housing data it is 13 weeks. Validation forecasts used to estimate fusion weights are restricted to targets dated on or before the training cutoff, so that no validation target falls inside the embargo. Test targets dated on or after January 1, 2026, the start of the reserved period, are excluded from the backtest; that period is used only for the holdout in Section 4.2.3.

Accuracy is reported at each horizon in $\mathcal{H} = \{H_1, \ldots, H_q\}$. Within a domain every agent uses the same set. The reference forecast, which is also the denominator of the skill score $s_{k,i} = 1 - e_{k,i}/e_{b,i}$, is the last observed value, $\hat{y}^{\text{naive}}_{t+h} = y_t$. For n pairs the primary metrics are mean absolute percentage error and root mean squared error [Hyndman and Koehler, 2006],

$$\text{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|, \qquad \text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}. \tag{12}$$

Accuracy differences are tested on the weekly cross-section of losses. For each pair of methods, the panel Diebold–Mariano test of equal average accuracy [Diebold and Mariano, 1995, Qu et al., 2024] uses a Newey–West variance with lag $H - 1$ [Newey and West, 1987]; per-MSA tests use the small-sample correction of Harvey et al. [1997] with the Benjamini–Hochberg adjustment across the 15 MSAs [Benjamini and Hochberg, 1995]. Confidence intervals come from a moving block bootstrap over whole weeks [Künsch, 1989] with 5,000 replications and a block length of 13 weeks. Two methods are called equivalent when the 90% confidence interval of their MAPE difference lies within $\pm 2\%$ of the reference method's MAPE. The model confidence set [Hansen et al., 2011] summarizes the comparison among all methods.

Each agent is also scored alone by $L(A_k, H)$, a horizon-specific accuracy loss for agent A_k, and by skill against the reference. Removing channel k from the fused forecast changes loss by $\Delta^{(H)}_k = L(\text{FACTS}_{-k}, H) - L(\text{FACTS}, H)$. A positive value means that the channel contributes inside the combination. Removing a channel renormalizes the remaining weights in each cell without re-estimating them. Removing adaptive weighting replaces every cell's weights with the fold's pooled weights, which removes both horizon and regime conditioning, so this ablation alone cannot separate the two. Domain-specific constants such as event-impact scales, event durations, and reversion caps are selected by the calibration procedure on historical validation folds, before the test interval begins. For a candidate θ_j, $\Delta_f(\theta_j) = L_f(\theta_j) - L_f(\theta_{\text{current}})$ is the change in fold-f validation loss; a negative value is an improvement, and promotion requires that improvement across folds. A null candidate is included when it is meaningful, so an unsupported adjustment can be set to zero.

Two weighting rules are compared on the same forecasts: one pooled weight vector per fold, and separate weights for each combination of regime and horizon. For each fold and horizon, the rule with lower

validation error is selected, $g^* = \arg\min_g L_{\text{val}}(g)$, and fixed before the test interval. The regime is computed at the forecast origin, and test error is also reported by horizon and regime. To check whether regime labels carry information, the same pipeline is rerun with regime labels permuted in blocks and with horizon-only weights that ignore the regime.

Every compared method is scored at the same forecast origins and against the same realized targets. The references are the last observed value, a seasonal-naive forecast, and the seasonally adjusted naive forecast of the M4 competition [Makridakis et al., 2020]. The statistical baselines are Holt's linear-trend model, its damped form, factor-adjusted exponential smoothing with and without damping, and a factor-adjusted Theta model [Assimakopoulos and Nikolopoulos, 2000]. The strongest single model chosen on validation data is also reported, together with equal-weight combinations of all channels and of the temporal and seasonal channels. The learned baselines are the TimesFM 2.5 foundation model and a chain-of-thought language-model forecast. The primary success check is that the fused FACTS-MAS forecast improves on the last-observed-value reference over the full test window; the stricter check is whether it improves on the strongest single model. Design choices that used test data during development are listed in the protocol paragraph above, and the final configuration, hypotheses, and tests were fixed before the holdout was accessed. Cross-domain transfer keeps the same architecture and learns a new evidence mapping for each domain (Section 5.4).

4.2 U.S. Housing Inventory Forecasting

The weekly for-sale housing inventory time series from Zillow is the first complete instantiation of FACTS-MAS. It can populate all five channels in Section 3.1: recent inventory dynamics, national financial conditions, annual listing seasonality, metropolitan structure, and disaster declarations. Two of these channels, the intrinsic and event channels, calibrate to a null quantitative effect and are retired from the final housing forecast; they are reported as tested channels. The task is to forecast weekly inventory for 15 U.S. metropolitan statistical areas at 4-, 8-, and 13-week horizons. What follows specifies the housing configuration, reports forecast accuracy, and then diagnoses which channels earned influence. This case study is one applied configuration of FACTS-MAS, not a definition of the framework: the same procedure can be reused with a different target and a different evidence map.

4.2.1 Domain Instantiation

The target variable is weekly for-sale housing inventory obtained from Zillow Research. The study includes the following 15 metropolitan areas:

New York	Los Angeles	Chicago
Dallas	Houston	Washington, DC
Miami	Philadelphia	Atlanta
Phoenix	Boston	San Francisco
Riverside	Detroit	Seattle

Zillow reports this series as a four-week trailing average; the unsmoothed weekly series is used as a robustness check. The panel begins on February 3, 2018. The backtest uses observations through February 21, 2026. The holdout uses a fresh download taken on October 20, 2026 with observations through October 10, 2026 (454 weekly observations per metropolitan statistical area (MSA) and 6,810 MSA–week observations); every value dated on or before February 21, 2026 in that download equals the earlier vintage, so no revision affects the results.

Nine target observations were missing in Zillow's source series: four in Chicago, two in Los Angeles, two in Miami, and one in Atlanta. These gaps were filled independently within each MSA using the most

Table 2: FACTS-MAS evidence-channel instantiation for housing inventory.

Channel	Housing evidence	Implementation
Temporal	Historical weekly inventory	4 weeks: ETS with additive trend and no seasonal component; 8 and 13 weeks: damped-trend ETS on seasonally adjusted inventory (factor-adjusted)
Exogenous	Mortgage rate, federal funds rate, CPI, unemployment	Lagged ridge-regression modifier with pre-specified lags
Seasonal	Week-of-year inventory pattern	Ratio-to-moving-average seasonal factors (seasonally adjusted naive form)
Intrinsic	Population, income, housing stock, density, hazard exposure	Scaled ridge regression with structural mean reversion; retired (calibrated reversion cap is zero)
Event	FEMA disaster declarations	LLM-assisted event interpretation with confidence gating; retired (calibrated impact scale is zero)

recent previously observed value. No information from a future observation or another MSA was used during imputation, and the affected rows were retained in a separate audit log. The unsmoothed series has 20 missing observations, filled the same way. No missing observation falls in a test or holdout target. Table 2 maps the domain-independent FACTS-MAS evidence roles to their housing-specific implementations. The initial decomposition is deliberately non-overlapping: the 4-week temporal agent does not model seasonality internally because seasonality is represented by a separate evidence channel, which makes the incremental value of each component easier to evaluate. At 8 and 13 weeks, however, the temporal agent that survived validation is a factor-adjusted model that reuses the seasonal channel's factors (see the temporal channel below), so the two channels are not independent at those horizons.

Temporal channel The temporal channel models the endogenous trajectory of each MSA's inventory series independently. At the 4-week horizon it uses exponential smoothing with an additive trend and no seasonal component [Holt, 2004, Winters, 1960, Hyndman et al., 2002, Pan, 2011], $\hat{\mathbf{y}}_{\mathrm{AR}} = \mathrm{ETS}(y_{1:t}; \mathrm{trend} = \mathrm{add}, \mathrm{seasonal} = \varnothing)$. At 8 and 13 weeks it uses a factor-adjusted damped-trend model: the series is divided by the seasonal factors of the seasonal channel, a damped additive trend [Gardner and McKenzie, 1985] is fitted to the adjusted series, and the forecast is multiplied by the factors of the target weeks. This variant was promoted by a rule fixed before the comparison was run: it had to improve validation error and improve test error in at least five of the six folds; it met the rule at 8 and 13 weeks but not at 4 weeks. Because this rule consults test-fold accuracy, the holdout in Section 4.2.3 provides the clean test of the resulting configuration. Damping was chosen over an undamped trend on validation data in every fold. Both models are fitted separately for each MSA and forecast origin using only the available historical target, and negative extrapolated values are clipped at zero. A last-value persistence forecast, $\hat{y}^{\mathrm{naive}}_{t+h} = y_t$, serves as the minimum baseline. Over all test origins, the undamped trend model has MAPE 2.368%, 5.430%, and 10.225% at 4, 8, and 13 weeks, against 4.194%, 7.225%, and 10.814% for the naive forecast, and it improves on the naive forecast in 15, 14, and 11 of the 15 MSAs. Its advantage therefore narrows sharply with horizon, and at 13 weeks it is worse than the naive forecast in four of the six folds. The factor-adjusted damped model has MAPE 1.974%, 3.608%, and 5.664%.

Exogenous channel The exogenous channel uses four national macroeconomic series from Federal Reserve Economic Data (FRED): the 30-year fixed mortgage rate, the effective federal funds rate, the Consumer Price Index, and the unemployment rate. The weekly macroeconomic panel begins in January 2018. Because several FRED series are indexed by their reference month rather than their actual publication date, the housing pipeline shifts monthly observations forward by approximate publication delays before they become eligible for forecasting: 32 days for the federal funds series, 35 days for unemployment, and 42 days for CPI. Weekly mortgage rates require no corresponding monthly-release adjustment. The macro and Zillow series retain their native Monday and Saturday conventions, respectively. Macro observations are attached to each Zillow week using a backward-only as-of join, ensuring that a housing observation never receives a macroeconomic value dated after that week. The Macro agent (exogenous channel) fits a ridge regression of inventory percentage change on lagged macroeconomic features [Hoerl and Kennard, 1970]. The lags are pre-specified from the literature rather than fitted: 6 weeks for the mortgage rate, 10 weeks for the federal funds rate, 10 weeks for CPI, and 5 weeks for unemployment. Importantly, subsequent validation did not produce convincing evidence that this magnitude channel adds incremental predictive information. Three increasingly restrictive analyses were performed: direct forecast-signal correlation, lag-sensitive regression analysis, and a two-way fixed-effects panel regression with MSA-clustered standard errors. The final panel specification contained 5,850 observations. Mortgage rate, the theoretically most direct housing-finance variable, was statistically indistinguishable from zero ($p = 0.575$); unemployment was also null ($p = 0.548$), and CPI was borderline ($p = 0.085$). The federal funds rate had $p = 0.042$, but this does not survive a Holm adjustment across the four variables ($p = 0.170$), a wild cluster bootstrap ($p = 0.073$), or Driscoll–Kraay standard errors that allow for the fact that all MSAs share the same national regressor [Driscoll and Kraay, 1998] ($p = 0.71$). Used as a forecast on its own, the channel is no more accurate than the naive forecast (MAPE 4.258%, 7.297%, and 10.919%). Because this pattern did not constitute robust evidence of incremental forecasting value, we treat the macro magnitude channel as a documented negative result rather than continuing to tune the specification until a positive result appears. This behavior illustrates a central FACTS-MAS principle: candidate evidence channels are evaluated empirically and are not guaranteed positive influence merely because they are economically plausible.

Seasonal channel The seasonal channel is deterministic and learns a separate annual inventory profile for each MSA. For observation t, let m_t denote a 52-week centered moving average. A raw seasonal ratio is $r_t = y_t/m_t$. For ISO week $w \in \{1, \ldots, 53\}$, the seasonal factor q_w is the median historical ratio observed for that week. The week-of-year profile is smoothed with a seven-week circular moving average, so that the transition between week 53 and week 1 is treated as adjacent rather than as a boundary. Factors are then normalized to have mean one. The window was selected on validation data, nested by fold, from $\{1, 3, 5, 7\}$; the value seven was chosen in every fold. An earlier window of five had been chosen on the test window of the last fold and is not used. Near the forecast origin the centered average is computed from the available half-window only, so no factor uses data after the origin. If q_{w_t} denotes the seasonal factor at the forecast origin and $q_{w_{t+h}}$ the factor for future week $t + h$, the seasonal forecast is $\hat{y}^{\text{season}}_{t+h} = y_t q_{w_{t+h}}/q_{w_t}$. This forecast is the seasonally adjusted naive method of the M4 competition [Makridakis et al., 2020] with leakage-safe factors. On its own it has MAPE 2.890%, 4.774%, and 6.945% at 4, 8, and 13 weeks.

Intrinsic channel The intrinsic channel represents persistent differences between metropolitan areas. The five static features are population, median income, housing units, density, and a hazard index. Population, income, and housing stock are derived from Census American Community Survey data. Population density combines Census population with CBSA land-area information. Hazard exposure is represented using county-level NOAA Storm Events records aggregated to each MSA. An initial tree-based implementation strongly overfit the small cross-section of 15 metropolitan areas. The current implementation

Table 3: Housing Event Agent impact-scale calibration: mean validation MAPE. Validation targets are restricted to dates on or before each fold's training cutoff; an earlier version of this sweep allowed targets to extend up to 13 weeks past it, which changed the values by 1–2% but not the ordering.

Scale	0	.01	.02	.03	.05	.075	.10
MAPE	**5.704**	5.869	6.112	6.413	7.110	8.121	9.234

therefore uses a standardized ridge regression to estimate a structurally implied average inventory level, $\mu_d = \beta_0 + \boldsymbol{\beta}^\top \tilde{\mathbf{x}}_d$. The model is regularized and evaluated using held-out MSAs so that structural relationships must generalize across metropolitan areas rather than memorize MSA identity. The current implementation replaced an earlier Gradient-Boosting model that achieved very low training error but approximately 48–52% leave-one-MSA-out error. Rather than immediately forecasting the structural level itself, the agent gradually moves the most recent inventory toward μ_d. For horizon H and step h, $\rho_h = \rho_{\max}(h/H)^2$ and $\hat{y}^{\text{intrinsic}}_{t+h} = y_t + \rho_h(\mu_d - y_t)$, with $\rho_{\max} = 0.30$ in the original housing configuration. The cap was then calibrated on validation folds over $\{0, 0.10, 0.20, 0.30, 0.45, 0.60, 0.80\}$, and zero was preferred in all six folds (validation MAPE 7.923% at zero against 10.259% at 0.30). At $\rho_{\max} = 0$ the agent reproduces the naive forecast, so the channel is retired from the housing forecast. The quadratic schedule is also not consistent across horizons: the forecast for week $t + 4$ closes 0.30 of the gap when $H = 4$ but 0.028 of it when $H = 13$. Because the channel is retired, this is documented rather than corrected. The static features use the 2024 Census vintage and 2016–2025 storm records for every fold; replacing the channel's output with the naive forecast changes fused accuracy by at most 0.011 percentage points, so this vintage has no material effect.

Event channel Discrete events are represented using FEMA disaster declarations. OpenFEMA provides stable declaration metadata, which is mapped from affected counties to the 15 target MSAs. The Event Agent is the only housing component that uses an LLM (gpt-4o-mini). Its role is not to directly generate an inventory forecast from free-form text. Instead, the model converts event metadata into a structured representation containing event type, impact magnitude, confidence, and reasoning. The event ontology includes disaster, zoning change, major employer shift, and housing initiative categories. Event confidence is explicitly gated: interpretations with confidence less than or equal to 0.60 receive zero quantitative effect. Both the original magnitude and gated result are retained for auditability. Prompt development revealed an important failure mode. Without an explicit causal definition of the magnitude sign, small prompt changes caused event impacts to reverse direction. The final interpretation procedure therefore requires the model to identify the mechanism connecting an event to for-sale inventory before assigning the sign of the impact. FEMA's available metadata nevertheless supported only coarse severity differentiation. In practice, the LLM largely behaved as an ordinal severity classifier rather than a genuinely continuous impact estimator. The resulting event-magnitude mapping was therefore subjected to the calibration procedure in Sections 3.6 and 4.1. Two event assumptions were explicitly calibrated: the duration for which a declaration should influence the forecast and the scale converting event severity into a numerical inventory adjustment. For the active-event duration, progressively shorter windows produced lower validation error in all six folds. Once the impact scale is set to zero (below), the duration no longer affects the forecast. A separate sweep of the event impact scale considered the values 0, 0.01, 0.02, 0.03, 0.05, 0.075, and 0.10. Mean validation MAPE increased monotonically as the event magnitude scale increased (Table 3). The scale of zero was preferred in 6 of 6 folds. With a zero scale, the agent's forecast equals the last observed value at every origin, so the event channel reproduces the naive reference and is retired from the final housing forecast. Consequently, the production Event Agent does not directly perturb forecast magnitude from the LLM severity score. Event

Table 4: Housing backtest folds and holdout. Test start and end are the first and last forecast origins.

Fold	Train start	Train end	Test start	Test end
1	2019-02-02	2021-01-23	2021-05-01	2021-11-13
2	2019-02-02	2021-11-13	2022-02-19	2022-09-03
3	2019-02-02	2022-09-03	2022-12-10	2023-06-24
4	2019-02-02	2023-06-24	2023-09-30	2024-04-13
5	2019-02-02	2024-04-13	2024-07-20	2025-02-01
6	2019-02-02	2025-02-01	2025-05-10	2025-11-22
Holdout	2019-02-02	2026-02-21	2026-02-28	2026-09-12

classification and confidence remain available, but the unsupported cardinal interpretation of the severity score is removed. This result should not be interpreted as evidence that real-world events are irrelevant to housing. It shows that the particular mapping from available FEMA metadata to a continuous point-forecast adjustment did not demonstrate incremental predictive value.

The housing experiment uses six expanding-window folds and reserves data from January 2026 onward. In an earlier version of the experiment, 8- and 13-week forecasts issued near the end of the sixth test window had targets in January and February 2026 (165 MSA–origin–horizon cells); these cells are excluded from every backtest result reported here. The fold schedule is shown in Table 4, together with the holdout described in Section 4.2.3. Every fold contains at least a 13-week separation between the final eligible training date and the test period, matching the longest housing forecast horizon. Forecasts are evaluated at every eligible weekly origin within each test window rather than at one forecast date per fold.

To evaluate whether agent usefulness changes with financial conditions, housing forecast origins are divided into three monetary-policy regimes using the 13-week change in the federal funds rate, $\Delta^{13}\mathrm{FFR}_t = \mathrm{FFR}_t - \mathrm{FFR}_{t-13}$. The origin is hiking when that change exceeds 0.25 percentage points, cutting when it is below -0.25, and stable otherwise. The regime is evaluated at the forecast origin rather than the target date, ensuring that it represents information available when the prediction is made. Because the federal funds rate is national, all 15 MSAs share the same regime at every date, and the sample contains one substantive hiking episode (June 2022 to September 2023) and three cutting episodes, so regime conditioning is close to conditioning on a small number of historical periods.

Two fusion configurations are compared: pooled fold-level weighting, and adaptive selection, for each horizon, between pooled weights and weights conditioned on regime and horizon. The production configuration uses adaptive selection with the 13-week data-sufficiency rule of Section 3.5. A finer, cell-level selection was also explored during development, but that choice was informed by test-window results and is not reported. Section 5.1.5 tests whether the regime labels carry information.

4.2.2 Forecasting Results

Table 5 reports test accuracy over the six backtest folds for FACTS-MAS, its references, and the statistical baselines, on both Zillow series. FACTS-MAS outperforms the persistence reference at all three horizons: the panel Diebold–Mariano statistic is -6.9, -6.3, and -5.7 at 4, 8, and 13 weeks ($p < 10^{-4}$), the difference is significant in all 15 MSAs at every horizon, and the 95% confidence intervals of the MAPE difference are $[-2.66, -1.42]$, $[-4.72, -2.52]$, and $[-7.11, -3.53]$. The relative reduction is 49.5%, 49.9%, and 47.0% on the smoothed series and 23.9%, 34.5%, and 37.1% on the unsmoothed series. The strongest single model, factor-adjusted damped exponential smoothing (FA-ETS-d), is at least as accurate as FACTS-MAS at every horizon. FACTS-MAS is statistically equivalent to it at 8 weeks (90% interval of the difference $[-0.001, 0.030]$, within the ± 0.072 margin), less accurate at 4 weeks by 0.144 (90% interval $[0.022, 0.261]$;

Table 5: Backtest MAPE (%) over six folds, all test origins, targets before January 1, 2026. FA-ETS-d: factor-adjusted damped ETS. Lower is better; bold marks the lowest value in each column.

	Smoothed series			Unsmoothed series		
Method	4 wk	8 wk	13 wk	4 wk	8 wk	13 wk
Naive (last value)	4.194	7.225	10.814	4.785	7.882	11.438
Seasonal naive	20.771	20.513	20.597	–	–	–
Seasonally adjusted naive	2.890	4.774	6.945	3.626	5.408	7.489
Holt (temporal channel)	2.368	5.430	10.225	4.317	7.901	13.022
Theta (factor-adjusted)	2.933	4.842	7.047	–	–	–
FA-ETS-d	**1.974**	**3.608**	**5.664**	**3.402**	**5.105**	**7.164**
FACTS-MAS	2.118	3.622	5.731	3.642	5.159	7.189

Table 6: Backtest MAPE (%) over six folds against the TimesFM 2.5 foundation model, smoothed series.

Method	4 weeks	8 weeks	13 weeks
Naive	4.194	7.225	10.814
TimesFM 2.5	2.361	4.860	8.269
FA-ETS-d	**1.974**	**3.608**	**5.664**
FACTS-MAS	2.118	3.622	5.731

Diebold–Mariano $p = 0.12$), and marginally less accurate at 13 weeks by 0.067 (95% interval $[0.001, 0.152]$; $p = 0.09$). At 8 weeks the 90% model confidence set contains only FA-ETS-d, FACTS-MAS, and the five-channel variant of FACTS-MAS; at 13 weeks it excludes only the naive forecast. The fused forecast therefore improves strongly on the references but does not improve on the best single model.

Table 6 compares FACTS-MAS with Google's TimesFM 2.5 time-series foundation model [Das et al., 2024], run zero-shot with a 52-week context at every test origin of all six folds. TimesFM improves on the naive reference at every horizon ($p < 10^{-4}$), but FACTS-MAS is more accurate than TimesFM at every horizon, by 0.243, 1.239, and 2.539 percentage points (Diebold–Mariano $p = 0.006$, 0.0001, and 0.0004; significant in 7, 10, and 10 of the 15 MSAs). FA-ETS-d is likewise more accurate than TimesFM at every horizon. The chain-of-thought language-model forecast is evaluated on the holdout only (Section 4.2.3), because the backtest period falls inside the training data of the language models available for the comparison.

Errors stratified by monetary-policy regime show no regime in which the final forecast fails. At the 13-week horizon,

Regime	Naive	FA-ETS-d	FACTS-MAS
hiking	8.059	5.530	5.552
cutting	13.043	6.091	6.090
stable	11.628	5.666	5.759

so FACTS-MAS improves on the naive reference in every regime, and FA-ETS-d, which uses no regime information, is equally accurate in every regime. An earlier version of the system appeared to lose to the naive reference in the hiking and cutting regimes and to be repaired by regime-aware weighting; that pattern came from an older seasonal channel and, in the cutting regime, from the equal-weight fallback described in Section 3.5, and it does not appear in the current configuration.

Table 7: Holdout MAPE (%), forecast origins February 28 to September 12, 2026, no publication lag. 13-week values are descriptive. CoT: chain-of-thought forecast (gpt-5.6-luna), run on the smoothed series only.

	Smoothed series			Unsmoothed series		
Method	4 wk	8 wk	13 wk	4 wk	8 wk	13 wk
Naive (last value)	3.122	5.445	8.438	3.355	5.306	7.925
Seasonal naive	8.534	8.534	8.550	8.593	8.601	8.622
Seasonally adjusted naive	1.492	2.377	3.393	2.053	2.677	**3.564**
TimesFM 2.5	1.410	2.602	4.457	2.464	3.689	5.355
CoT (gpt-5.6-luna)	1.259	2.225	3.268	–	–	–
FA-ETS-d	**1.138**	1.936	3.080	**2.028**	2.709	3.736
FACTS-MAS	1.243	**1.929**	**3.045**	2.053	**2.671**	3.644

4.2.3 Holdout Evaluation

After the configuration was frozen, it was evaluated once on data that had not been accessed during development: forecast origins from February 28 to September 12, 2026 (29, 25, and 20 weekly origins at 4, 8, and 13 weeks for each of the 15 MSAs). The fusion weights were estimated from the 104-week validation window ending on February 21, 2026. Four hypotheses, the tests, and the reporting rules were committed before the holdout data were downloaded: FACTS-MAS improves on the naive reference at every horizon (H1); FA-ETS-d does too (H2); FACTS-MAS and FA-ETS-d are equivalent at 8 and 13 weeks (H3); and FA-ETS-d is at least as accurate as FACTS-MAS at 4 weeks (H4). With 20 origins the 13-week results are reported as descriptive only. The bootstrap block length equals the horizon, and at 13 weeks it is capped at half the number of weeks. The chain-of-thought baseline uses the NEXUS prompt with gpt-5.6-luna (reasoning effort low), whose documented knowledge cutoff, February 16, 2026, precedes the first holdout target; the model's own answer to a cutoff question was June 2024. The prompt itself states a January 2025 cutoff, a wording inherited from the NEXUS template.

Table 7 reports the holdout accuracy and Table 8 the hypothesis tests. All four hypotheses are supported. FACTS-MAS reduces MAPE relative to the naive forecast by 60%, 65%, and 64% on the smoothed series and by 39%, 50%, and 54% on the unsmoothed series. It is equivalent to FA-ETS-d at 8 weeks and less accurate than FA-ETS-d at 4 weeks by 0.105 percentage points. The chain-of-thought forecast is competitive: its MAPE is higher than that of FA-ETS-d and of FACTS-MAS at every horizon, but neither difference is statistically significant (against FA-ETS-d, $p = 0.38$, 0.43, and 0.60; against FACTS-MAS, in a comparison added after the results were seen, $p = 0.90$, 0.43, and 0.54). TimesFM is significantly less accurate than FA-ETS-d at 4 and 8 weeks ($p = 0.016$ and $p < 10^{-4}$). On the unsmoothed series FACTS-MAS at 4 weeks coincides with the seasonally adjusted naive forecast, and at 8 weeks it is slightly more accurate than FA-ETS-d (-0.038, $p = 0.031$); because the seasonally adjusted naive forecast is equally accurate there, this does not indicate a benefit of fusion over a single model. With a two-week publication lag, every method's error rises (FACTS-MAS 2.270%, 2.944%, and 4.000%; naive 5.474%, 7.953%, and 11.110%), and no conclusion changes. In the lag-two run, 27 of 1,110 chain-of-thought forecasts (2.4%) were lost to a network interruption; the tests for that run were computed with and without the chain-of-thought forecast. The holdout period was easier for every method than the backtest, so absolute errors should be compared within a period, not across periods.

Several limits are inherited from the housing data. FRED publication adjustments use approximate fixed delays rather than true vintages: 32 days for federal funds, 35 for unemployment, and 42 for CPI. That removes the most obvious lookahead but does not reconstruct the exact information set on every historical date. The target is Zillow's smoothed series, a four-week trailing average, which makes every method look

Table 8: Pre-registered holdout hypotheses, smoothed series, no publication lag. Differences are in MAPE percentage points with 95% block-bootstrap intervals (90% intervals for the equivalence test, H3).

Hypothesis	Horizon	Difference	Interval	Result
H1: FACTS-MAS $<$ naive	4 wk	-1.879	$[-3.031, -0.897]$	supported
	8 wk	-3.516	$[-5.681, -1.705]$	supported
	13 wk	-5.393	$[-7.681, -3.071]$	supported
H2: FA-ETS-d $<$ naive	4 wk	-1.984	$[-3.081, -1.031]$	supported
	8 wk	-3.509	$[-5.693, -1.680]$	supported
	13 wk	-5.358	$[-7.642, -3.028]$	supported
H3: equivalent to FA-ETS-d	8 wk	-0.007	$[-0.029, 0.014]$	equivalent (± 0.039)
	13 wk	-0.035	$[-0.048, -0.035]$	equivalent (± 0.062)
H4: FA-ETS-d $\leq$ FACTS-MAS	4 wk	$+0.105$	$[-0.007, 0.182]$	supported

Table 9: FACTS-MAS evidence-channel mapping for unemployment insurance claims. The temporal, exogenous, and seasonal channels are implemented; the intrinsic and event channels are not.

Channel	Unemployment-domain interpretation
Temporal	Historical unemployment level, rate, or claims trajectory
Exogenous	Macroeconomic, labor-market, financial, or activity indicators
Seasonal	Recurring calendar structure in employment or claims
Intrinsic	Persistent regional labor-force, demographic, or industrial composition
Event	Major layoffs, employer expansions or closures, policy changes, disasters, or other labor-market shocks

more accurate than on the unsmoothed series; both are reported. The hazard index is a partial structural proxy: NOAA Storm Events can be coded by county or by forecast zone, and the present aggregation uses county-coded events, so some hazards are undercounted. The Intrinsic Agent learns mainly from differences across metropolitan areas, and only 15 MSAs are available. That small cross-section limits how much structural signal can be estimated. Replacing an overfit tree with ridge regression reduces memorization of those few areas, but it does not increase the number of metropolitan areas available for learning. Nine Zillow observations were forward-filled, including a four-week Chicago gap. The fill is lookahead-safe, yet it inserts an artificial flat stretch.

The Event Agent currently depends on FEMA declarations. The event ontology used by this agent includes four categories—disaster, zoning change, major employer shift, and housing initiative—and therefore contemplates local shocks such as employer moves, mass layoffs, zoning actions, and housing-policy changes. FEMA covers only disasters, not the full set. Some FEMA records also fit the ontology poorly; a presidential-inauguration declaration is the example encountered in validation. Confidence gating zeroed its quantitative effect, but the gap remains. Two design choices made during development used test-window evidence (the original seasonal smoothing window and a cell-level fusion rule); both were replaced or removed, and the holdout was evaluated only after the configuration was frozen.

4.3 Unemployment Forecasting

Unemployment forecasting emphasizes labor-market dynamics, policy and employer shocks, and recurring calendar structure in employment or claims. The channel map for this domain is Table 9.

The target is weekly initial unemployment insurance claims, not seasonally adjusted, for 15 U.S. states

Table 10: Unemployment insurance claims: test MAPE (%) over six folds, pandemic window excluded. The undamped FA-ETS row was added after the other results were seen and is descriptive.

Method	4 weeks	8 weeks	13 weeks
Naive	18.816	21.963	24.716
Seasonally adjusted naive	17.836	19.294	20.613
FA-ETS-d (damped)	17.868	19.411	20.791
FA-ETS (undamped)	**17.71**	**19.18**	**20.55**
FACTS-MAS (frozen recipe)	17.863	19.233	20.578

from January 2013, obtained from FRED. The temporal and seasonal channels use the housing implementations unchanged, including the seven-week seasonal window, and the exogenous channel uses the federal funds rate and CPI with the housing lags. The housing recipe is applied without re-tuning: the same six folds, embargo, validation window, data-sufficiency rule, temperature, and promotion outcome. The window of pandemic-era claims, from March 14, 2020 to September 4, 2021 (the start of the national surge to the expiry of the federal pandemic programs), is excluded from skill estimation and scoring but kept for model fitting. This window, and the rest of the protocol, were fixed from policy dates and written down before any result of this evaluation was computed.

Table 10 reports the results. The fused forecast improves on the naive reference at every horizon ($p = 0.002$, 0.0004, and 0.0004) and on damped FA-ETS at 8 and 13 weeks ($p = 0.0009$ and 0.005), but validation selected the undamped trend in every fold, and the undamped factor-adjusted model alone is slightly more accurate than the fused forecast at every horizon. In unemployment claims, the frozen recipe beats naive and the damped FA-ETS reference, but not undamped FA-ETS, which is the better reference for that series. In neither domain is there evidence that fusion beats the best single model. The channel weights differ from housing, as the transfer hypothesis expects: with the temporal channel that ignores seasonality, the seasonal channel receives about three quarters of the weight at every horizon, and the exchange of weight between the temporal and seasonal channels seen across horizons in housing does not appear. Including the pandemic window raises every method's MAPE by 0.7 to 2.2 points. Because an earlier pilot evaluation scored claims data through August 2026, a clean holdout for this domain is not available.

4.4 Energy Forecasting

Energy demand forecasting emphasizes short and medium cycles, weather sensitivity, and abrupt regime changes driven by temperature extremes, outages, and price shocks. The intended channel map for this domain is Table 11. No energy experiments are reported in this paper; the mapping below records the planned configuration.

The configuration is $\mathcal{C}^{(\text{energy})} = (\mathcal{E}^{(\text{energy})}, \mathcal{A}^{(\text{energy})}, \Theta^{(\text{energy})}, \mathcal{H}^{(\text{energy})}, \mathcal{R}^{(\text{energy})})$. Completed energy runs should report target, frequency, regions, coverage, horizons, leakage controls, channel models, calibrated parameters, baselines, and any regime definition, together with learned weights $\mathbf{w}^{(\text{energy})} = [w_{\text{temporal}}, w_{\text{exogenous}}, w_{\text{seasonal}}, w_{\text{intrinsic}}, w_{\text{event}}]$, or the subset of channels actually implemented. In this domain, seasonal and exogenous channels are expected candidates for substantial weight when validation supports that attribution.

5 Discussion

The housing inventory case study shows that FACTS-MAS can decompose mixed evidence into testable channels, tell those channels apart, calibrate domain assumptions, and adapt their influence. It also clarifies

Table 11: Planned FACTS-MAS evidence-channel mapping for the energy domain.

Channel	Energy-domain interpretation
Temporal	Historical energy-demand or load trajectory
Exogenous	Weather, temperature, economic activity, price, or other external demand drivers available at the forecast origin
Seasonal	Recurring daily, weekly, and/or annual demand structure
Intrinsic	Persistent characteristics of the forecasting region, grid, customer base, or system capacity
Event	Discrete weather, outage, policy, infrastructure, or other domain-relevant events

what "domain-agnostic" does and does not mean. The framework is not a pretrained model whose fitted weights are expected to carry over unchanged to a new domain; instead, it carries over a procedure: mapping evidence to functional channels and scoring each channel against a reference, so as to calibrate, attribute, and fuse.

5.1 Channel Diagnostics and Fusion Analysis

This section examines standalone channel value, domain-specific calibration, and adaptive fusion: whether FACTS-MAS can distinguish useful evidence from weak, redundant, or context-dependent evidence and adjust the fused forecast accordingly.

5.1.1 Evidence Channels as Testable Hypotheses

Each FACTS-MAS agent is a forecasting hypothesis: $\mathcal{H}_k$ asserts that E_k contains incremental information about future y. Including an agent does not accept $\mathcal{H}_k$; historical validation decides whether the channel beats the reference. Unlike a joint feature model, FACTS-MAS produces independent agent forecasts before fusion, so weak evidence can be diagnosed without removing or retraining unrelated channels. The housing experiments show that pattern clearly. The temporal channel improves on persistence at 4 weeks but its advantage narrows with horizon unless seasonality is removed first; seasonality helps at every horizon; the tested macroeconomic magnitude channel adds little robust incremental value; the intrinsic reversion calibrates to zero; and the initial event-severity magnitude mapping is harmful before calibration. Agents are therefore candidate hypotheses, not mandatory contributors.

5.1.2 Macro Channel: A Well-Earned Null

The Macro agent (exogenous channel) is driven by mortgage rates, monetary policy, inflation, and unemployment. These are plausible housing drivers, but plausibility does not establish short-horizon incremental forecast value. Three diagnostics were used.

Direct forecast-signal validation. The agent's predicted modifier was compared with realized inventory changes across the six backtest folds. The correlations were weak:

Horizon	4 wk	8 wk	13 wk
r	0.011	-0.025	-0.177
p	0.920	0.814	0.095

(13)

providing no clear predictive signal.

Table 12: Final housing Macro agent signal validation: p-values for the original panel (MSA and year–quarter fixed effects, MSA-clustered errors), with a Holm adjustment across the four variables, a wild cluster bootstrap (Webb weights, 9,999 draws), and Driscoll–Kraay errors with MSA fixed effects. $n = 5{,}850$.

Variable	Lag	Coef.	Clustered	Holm	Wild bootstrap	Driscoll–Kraay
Mortgage rate	6 weeks	0.0075	0.575	1.000	0.611	0.973
Federal funds	10 weeks	-0.0129	0.042	0.170	0.073	0.712
CPI	10 weeks	-0.0047	0.085	0.254	0.086	0.463
Unemployment	5 weeks	-0.0015	0.548	1.000	0.577	0.264

Lag-sensitive analysis. A search over plausible transmission lags yielded apparently strong correlations for several raw macroeconomic levels. Those correlations are misleading: national macro values are identical across MSAs on a given date, so MSA–date pairs pseudo-replicate one national signal and overstate the effective sample size. Strong trends in both macro levels and inventory also make simple level correlations vulnerable to spurious association.

Panel validation. The final diagnostic used a two-way fixed-effects panel with MSA and year–quarter fixed effects, MSA-clustered standard errors, 5,850 observations, and the pre-specified midpoint lags rather than data-mined best lags (Table 12).

Only federal funds crosses the conventional 5% threshold under clustered errors, and it does not survive the multiple-testing adjustment, the wild bootstrap, or errors that allow for the common national regressor; mortgage rates, the most direct theoretical housing-finance channel, are null under every specification. We treat this as a *well-earned null*: the Macro agent was not retuned until a positive result appeared, and its weak incremental skill is allowed to reduce its fusion weight. FACTS-MAS can reject a plausible channel without redesigning the architecture.

5.1.3 Event Channel: Semantic Evidence and a Null Scale

The Event Agent fails differently. FEMA metadata contain useful semantic information that an LLM can classify, but a qualitative severity score is not automatically a valid continuous forecast adjustment. The original agent returned an impact in $[-1, 1]$. Prompt development showed that FEMA metadata support only coarse severity tiers: similar assistance-program configurations repeatedly mapped to the same severity, so the output behaved more like an ordinal classifier than a continuous impact model. FACTS-MAS therefore tested whether those magnitudes should modify the numerical forecast at all.

Impact-scale calibration. Let m_t be the signed severity score and λ the scale that maps it into a forecast adjustment. Instead of assuming $\lambda > 0$, calibration tested $\lambda \in \{0, 0.01, 0.02, 0.03, 0.05, 0.075, 0.10\}$ (Table 3).

The ordering is monotonic: larger magnitude adjustments worsen validation MAPE, and $\lambda = 0$ is preferred in all six folds. With the scale at zero, the declaration-active window no longer affects the forecast, and the event channel's output equals the naive reference. The production Event Agent therefore keeps semantic classification and confidence but removes the unsupported point-forecast magnitude adjustment. Useful semantic information need not imply a validated numerical effect.

5.1.4 From Raw Error to Softmax Weights

Why raw accuracy is insufficient. An earlier fusion rule weighted agents by inverse validation MAPE. The Macro agent, despite its null signal, sometimes received weight comparable to stronger agents. A weakly fitted regularized model can stay close to persistence; because persistence is competitive at short horizons, absolute error can look good without adding new information. Absolute error therefore confounds useful signal with naive imitation. FACTS-MAS uses skill relative to naive, $s_{k,i} = 1 - e_{k,i}/e_{\text{naive},i}$, so an agent that only reproduces the naive forecast receives approximately zero skill even when its MAPE looks competitive.

Robust skill aggregation. Averaging pointwise relative-skill ratios is unstable when $e_{\text{naive},i} \to 0$: the ratio $e_{k,i}/e_{\text{naive},i}$ can explode. In one diagnostic, the temporal agent had median pointwise skill about $+0.227$ but mean skill about -0.407, with isolated ratios near -32. Fusion therefore uses $\tilde{s}_k = \text{median}_i(s_{k,i})$. The median protects relative-skill aggregation whenever the reference occasionally has near-zero error.

Why softmax replaced hard skill floors. Transforming skill by $\max(s_k, \epsilon)^p$ avoided zero weights but collapsed every negative-skill agent to the same floor, so $s = -0.01$ and $s = -0.50$ became indistinguishable. Softmax, $w_k = \exp(\tilde{s}_k/\tau)/\sum_j \exp(\tilde{s}_j/\tau)$, preserves the full ranking, including among negative-skill agents. The fusion rule therefore evolved from raw MAPE, to relative skill, to robust relative skill with softmax, each step answering an observed failure of the previous rule.

5.1.5 Adaptive Fusion and Its Limits

Does regime conditioning carry information? Adaptive weighting chooses, for each fold and horizon, between pooled weights and weights conditioned on regime and horizon. Its apparent 13-week benefit was tested in three ways on the five-channel configuration, before the temporal channel was replaced. First, with regime labels permuted in blocks, 46% of 200 permutations at 8 weeks and 27% at 13 weeks were at least as accurate as the true labels. Second, weights conditioned on horizon alone, with no regime, matched the regime-conditioned weights at 8 weeks (4.468% against 4.468%) and were more accurate at 13 weeks (6.826% against 6.987%). Third, 60–75% of the 13-week benefit came from a single fold, in the stable regime. We therefore find no evidence that the regime labels carry predictive information beyond horizon conditioning; with a national regime variable and one substantive hiking episode in the sample, the regime partition is close to a partition by historical period.

Data sufficiency. Context-aware adaptation should depend on both estimated performance and evidence volume. Let $n_{H,r}^{(f)}$ be the number of distinct validation weeks for horizon H, regime r, and fold f. The production rule uses regime-conditioned weights only if $n_{H,r}^{(f)} \geq 13$ and otherwise falls back to the pooled weights. An earlier implementation had no such rule: when a regime was absent from a validation window, every channel, including the channels that reproduce the naive forecast, received a weight of 0.2. That fallback affected 780 test cells, accounted for about one fifth of the apparent 13-week benefit of regime weighting, and produced the apparent repair of the cutting regime reported in an earlier version of this work.

Fusion versus the strongest single model. Once the temporal channel is seasonally adjusted, validation gives it 0.97 of the weight at 8 weeks and 0.92 at 13 weeks, the seasonal channel receives most of the remainder, and the exogenous channel receives almost none. The fused forecast is then statistically equivalent to the factor-adjusted model alone at 8 weeks in the backtest and on the holdout, and slightly less accurate at 4 and 13 weeks, in the backtest and on the holdout. Three further candidates, each registered before it

was run, did not change this: a listing-flow channel built from Zillow new listings, newly pending sales, and price-cut shares; fusion of deviations from the naive forecast; and a validation-selected temperature with shrinkage toward equal weights. None met the promotion rule, and the best was 0.002 percentage points behind FA-ETS-d at 4 weeks ($p = 0.42$). In the housing domain, the value of the fusion step is therefore that it selects and recomposes the useful channels, not that it combines them into a forecast more accurate than the best of them.

5.1.6 Predictive versus Causal Attribution

Fusion weights measure predictive importance, not causation. For agent k, w_k is the influence assigned to its forecast given historical validation skill; it does not claim that the underlying evidence caused the same share of the target change. A low weight can mean little forecasting signal, redundancy with another channel, a weak implementation, rare-regime dependence, or insufficient validation history. Accordingly, *factor attribution* means predictive attribution within the forecasting architecture, not structural causal identification.

5.2 Why Full-Window Evaluation Matters

The baseline experiments also exposed a significant evaluation artifact. An earlier diagnostic evaluated one forecast origin per fold and horizon, and its comparison of the temporal channel with the naive forecast scored horizons that fell inside the embargo. Under that design the 13-week forecast appeared to lose to naive in most MSAs. A single forecast date is a poor estimator of general forecasting performance, so every result reported here uses all eligible forecast origins in each test window: 2,610 MSA–origin pairs per horizon in the backtest and 29 weekly origins per MSA in the holdout.

5.3 What the Housing Results Imply

FACTS-MAS improves on persistence at every evaluated horizon, on both Zillow series, in the backtest and on the holdout, and it is more accurate than the TimesFM foundation model. On the holdout a chain-of-thought language-model forecast comes close to it, with no statistically significant difference. It is not more accurate than the strongest single model, factor-adjusted damped exponential smoothing, which it matches at 8 weeks. The more important architectural point is that the final forecast is the product of tests that each channel had to pass. Temporal dynamics and seasonality carry skill; the tested macro magnitude specification is largely null; the intrinsic reversion and the event-magnitude map calibrate to zero. That pattern is intentional. Available evidence is not the same as validated predictive evidence: the first determines which agents can be proposed, and the second determines how strongly they may move the forecast. A less disciplined process could keep searching lags and prompts until a favorable statistic appears; FACTS-MAS treats a null channel as an admissible outcome.

The Event Agent (event channel) and Macro Agent (exogenous channel) illustrate two different limits of that distinction. An LLM can turn disaster metadata into structured categories, explanations, and confidence, but that does not imply a calibrated continuous effect; usefulness can stop at interpretation, classification, ordinal severity, or only at a cardinal forecast effect. The Macro agent fails for another reason: not semantic ambiguity, but too little independent variation. National macro series are repeated across MSAs, so naive correlations overstate the sample, and a panel with thousands of entity–time rows can still contain only a modest number of macro shocks.

Fusion weights should be read in the same spirit. For agent k, w_k is a predictive attribution, not a causal contribution: a large temporal weight does not mean that temporal dynamics caused the same share of the observed movement, and a small macro weight does not mean that monetary conditions are causally

irrelevant. Low weight can also mean redundancy, a weak implementation, rare-regime dependence, or insufficient validation history. Regime analysis shows why apparent context effects need a placebo. An earlier version of the system seemed to fail in the hiking and cutting regimes and to be repaired by regime-aware weighting, but the failure disappeared once the seasonal channel was estimated properly, the repair was partly an artifact of an equal-weight fallback, and randomly permuted regime labels did as well as the true ones. Context-specific weighting reduces the observations behind each decision, and the rare regimes where adaptation looks most attractive often have the least support. The tradeoff is context specificity versus estimation reliability. Energy or unemployment applications will meet the same constraint whenever extreme weather or recessions are uncommon in the validation sample.

5.4 Toward Cross-Domain Transfer

Housing shows that the procedure works inside one heterogeneous problem. It does not, by itself, show that the procedure is domain agnostic. Transfer in FACTS-MAS is reconfiguration of evidence, not identity of fitted parts.

For source domain $\mathcal{D}_1$ and target domain $\mathcal{D}_2$, we do not require $\mathcal{E}^{(\mathcal{D}_1)} = \mathcal{E}^{(\mathcal{D}_2)}$, nor $\mathbf{w}^{(\mathcal{D}_1)} = \mathbf{w}^{(\mathcal{D}_2)}$. Identical weights across unlike targets would be a warning that fusion is insensitive to domain structure. What should be preserved is the procedure: decompose the evidence, forecast from each channel, validate, calibrate, attribute, and fuse. Formally, each domain supplies a configuration $\mathcal{C}^{(\mathcal{D})} = (\mathcal{E}^{(\mathcal{D})}, \mathcal{A}^{(\mathcal{D})}, \Theta^{(\mathcal{D})}, \mathcal{H}^{(\mathcal{D})}, \mathcal{R}^{(\mathcal{D})})$, where $\mathcal{E}$ is the evidence map, $\mathcal{A}$ the channel models, Θ the calibrated parameters, $\mathcal{H}$ the horizons, and $\mathcal{R}$ optional regimes. The invariant pieces remain the shared interface, origin-time information boundary, leakage-safe rolling evaluation, reference-relative skill, calibration, adaptive fusion, and predictive attribution. Weights are re-estimated in the target domain: $\mathbf{w}^{(\mathcal{D}_1)} \neq \mathbf{w}^{(\mathcal{D}_2)}$ is expected rather than treated as failed transfer. The transfer hypotheses are therefore architectural, not parametric.

T1: Architectural transfer. A new problem can be written in the same functional channels and contracts without a new orchestrator.

T2: Adaptive predictive value. Channel importance changes across domains and is learned from each domain's own validation history.

T3: Forecasting benefit. After that reconfiguration, adaptive fusion should remain competitive with the new domain's simple reference and strong standalone models.

The unemployment-claims study (Section 4.3) supports T1: the claims series was written in the same channels and contracts, and the housing code ran unchanged. It supports T2: the learned weights differ from housing, with the seasonal channel carrying most of the weight. It supports T3 only in part: the fused forecast beats the naive reference and the damped factor-adjusted model, but not the undamped factor-adjusted model, which validation selected and which is the better single model for that series. The energy configuration in Section 4.4 is planned and untested.

6 Conclusion

This paper introduced FACTS-MAS, a multi-agent forecasting framework that treats heterogeneous evidence as a set of testable hypotheses rather than as a single pooled feature list. Each functional channel produces an independent forecast; leakage-safe validation measures incremental skill against a reference; calibration is allowed to return a null effect; adaptive fusion converts that evidence into weights that can depend on horizon and regime when the sample can support the dependence.

On weekly U.S. housing inventory across 15 metropolitan areas, the frozen configuration attains backtest MAPE of 2.118%, 3.622%, and 5.731% at 4, 8, and 13 weeks on Zillow's smoothed series, against 4.194%, 7.225%, and 10.814% for last-value persistence, and 1.243%, 1.929%, and 3.045% on a pre-registered holdout, against 3.122%, 5.445%, and 8.438%. The gains are smaller but still significant on the unsmoothed series. FACTS-MAS is more accurate than the TimesFM 2.5 foundation model; on the holdout its error is lower than that of a chain-of-thought language-model forecast built on the NEXUS prompt, but not significantly so. It is not more accurate than a factor-adjusted damped exponential smoothing model, with which it is equivalent at 8 weeks. Those results coexist with null and rejected channels: the tested macro magnitude specification adds little short-horizon skill, validation removes the intrinsic reversion and the cardinal event-severity adjustment, and regime-aware weighting shows no benefit beyond horizon conditioning once a placebo is applied. The contribution of FACTS-MAS is therefore the protocol itself: every channel is tested against a reference, calibration may return a null, the final forecast is recomposed from the channels that survive, and the result is checked on data held out until the configuration is frozen. Transferring the method to a different domain requires a new evidence map, new channel models where needed, and new validation-based calibration and weights, but the FACTS-MAS architecture, the shared interface, leakage-safe scoring, and adaptive fusion procedure remain the same.

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